

NODE Backbone

8-Channel Addressable LED Controller · Ethernet + PoE Node

Watefy Electronics

Wired control. The network powers the node.

At a Glance

- **8 independent output channels** — WS281x / SK6812 / TM1814 family, 5 V level-shifted drive
- **Wired Ethernet on dedicated silicon** — W5500 controller with its own MAC, PHY and hardware TCP/IP stack
- **Power over Ethernet** — the switch powers the controller; no separate supply for the electronics
- **5-48 VDC pixel input, no jumpers** — separate rail for strip power, nothing to set wrong
- **30 A through the board at standard copper** — not a paid upgrade
- **Three isolated pixel power domains** plus an independent network-powered logic domain
- **A fuse on every single output** — eight channels, eight fuses, no sharing
- **Four-port DMX-512 / RS-485 output** — per-port transient protection, one port receive-capable
- 123 × 129 mm · 4-layer · ENIG · conformal coated

Why This Exists

Wi-Fi is fine until it isn't. Put twelve controllers on one property, push sixty frames a second of sACN at all of them, and the failure mode is not a clean disconnect — it is stutter, tearing, and one zone that goes dark for four seconds during the show. Adding access points helps until the 2.4 GHz band gives up.

NODE Backbone puts the control plane on wire. Each node gets an Ethernet drop; the switch powers the logic and carries the data. Frame timing becomes deterministic, the controller shows up in your DHCP table like every other piece of infrastructure, and troubleshooting becomes a link light instead of a signal-strength guess.

This is the SKU for the installation that outgrew Wi-Fi.

The Architecture: Three Power Domains

Most controllers have one power rail. Input voltage lands on a terminal and it is the same copper that reaches your strips. Whatever arrives — spike, reverse polarity, sag, transient — arrives everywhere at once.

NODE splits power into three separate domains on their own copper planes:

- 1 • Dirty power in.** The input terminal accepts whatever the supply, the wiring, and the person doing the wiring hand it. This domain is assumed hostile, isolated to the smallest region of the board possible, and nothing downstream touches it directly.
- 2 • Clean power.** Between dirty and clean sits the protection stage: bidirectional transient clamping and an active reverse-polarity blocker. Reverse the supply and the pass element never turns on — no current path to the rest of the board, nothing sacrificed, red indicator tells you why. Correct the wiring and it recovers on its own.
- 3 • The protected bus.** Clean power passes the main fuse to become the distribution bus. Each of the eight output channels taps that bus through its own fuse. A fault on run six opens one fuse and the other seven never notice.

And a fourth domain, entirely separate: the network side. Logic power arrives over Ethernet, on its own ground domain, joined to board ground at one deliberate point. The control plane and the high-current pixel return are treated as a design problem rather than an accident of layout — the difference between a board that passes on a bench and one that behaves across long runs and multiple supplies.

BEEFCAKE. The pass element is an 80 V MOSFET whose current rating is several times the board's 30 A design load. It will never be the thing that fails.

Decoupling where the current actually is. Two thousand microfarads of bulk sits on the protected bus, and another forty-seven at every single output connector. The distribution is the point: one reservoir at the input cannot cover the harness inductance of eight separate output runs no matter how large it is. When a full-white frame lands on all eight channels at once, each channel draws from its own reserve sitting centimetres from its terminal.

Four levels between the screw terminal and your pixels — clamp, active block, main fuse, channel fuse — each on its own copper. That is why this is a four-layer board with a split inner power plane rather than two layers and one pour.

Nothing in the pixel power path is rated below **63 V** — and every semiconductor in it is **60 V or above** — against a 48 V absolute maximum input. Nothing in the chain runs near its limit.

At the **36 V nominal** rating, no component in the power path operates above **60 %** of its own rating. At the **48 V absolute maximum**, nothing exceeds **80 %**. On the 5 V, 12 V and 24 V strips that carry the overwhelming majority of installations, the entire plane runs under **40 %**. It is not a number anyone else in this class publishes, and it is the reason to believe the rest of this page.

What Else You Actually Get

A real network node. Wired 10/100 Ethernet with PoE means the logic side needs nothing but the drop. Run pixel power to the enclosure on conductors sized for the load and let the network cable handle control and controller power. Fewer supplies, fewer failure points, and a control path that behaves the same at node twelve as at node one.

Reachable even when the pixel supply is off. Because logic power comes over Ethernet, the node stays on the network with the strip supply dark. Useful for pre-season configuration and for telling "controller down" apart from "supply down" without a ladder.

Ethernet on dedicated silicon. The ESP32-S3 has no built-in Ethernet MAC, so this variant runs a W5500 — a standalone controller carrying its own MAC, its own PHY, and a hardware TCP/IP stack, talking to the MCU over SPI. The alternative in this class is an ESP32-classic driving an external PHY over RMII, which consumes nine GPIO and runs the network stack on the same cores rendering your effects. Here, the network is somebody else's job.

Data lines treated like data lines. Series resistance and a sub-picofarad ESD device per channel, placed on the connector side where transients arrive, chosen so the clamp does not round off WS281x bit timing on long runs.

DMX where you need it. Four RS-485 ports on RJ45, each with its own transceiver and its own transient protection, alongside the eight pixel channels.

Three things on this board can be killed by field abuse. All three pull out by hand.

The fuse. A mini blade — the same part sold in a variety pack at every gas station, auto parts counter and big-box store in the country. Pulls with fingers, costs pennies. Not a cartridge you order online and wait for.

The buffer. A jellybean octal logic IC in a DIP socket. It sits directly behind your wiring, which makes it the part most likely to die and the part most annoying to replace anywhere else. Here it lifts out with a fingernail. And the socket accepts either of two common octal logic families — a strap on pin 1 handles the output-enable behaviour of one and the direction pin of the other — so you fit whichever is in stock. Supply-chain resilience on a fifty-cent line item sounds trivial right up until the next shortage.

The MCU. A socketed module. Blow it up, seat another. No hot air, no braid, no scrap board. Nobody else in this class can claim all three.

And that is what the board area buys. Blade holders, a DIP socket, and terminal clearance cost real estate — there is room to get an iron in, room to pull a fuse without tweezers, room to reach the reset button without unmounting anything, and room at the terminals for heavy wire with ferrules. Compact is easy. Compact *and serviceable* at 30 A is a different problem, and this is the side of it we chose. A channel that dies at eight o'clock on the twenty-third of December should be a fifteen-minute drive, not a three-day wait.

Where It Goes

- **Large distributed displays** — a dozen or more nodes where Wi-Fi contention becomes the limiting factor
- **Permanent commercial and architectural installations** — where the lighting belongs in the building's network documentation
- **Long runs to outbuildings** — an Ethernet drop instead of a wireless prayer; pixel power still runs on its own conductors
- **High frame-rate sACN or Art-Net** — deterministic delivery, no shared-medium contention

How It Compares

Comparison basis. The closest peer is the Bong69 8 Port LED Distro — same idea, power and data integrated on one board, eight ports, wired Ethernet, WLED, sold at maker scale. QuinLED's Dig-Quad supports an Ethernet-capable brain board and is the other reference point. Competitor figures are taken from the vendors' own published FAQ and specification pages as of July 2026 and should be re-verified before publication.

Athom's WLED range is deliberately excluded: single-strip and PWM controllers aimed at room-level and audio-reactive use. Different product, different buyer.

	NODE Backbone	Bong69 8 Port LED Distro	QuinLED-Dig-Quad v3
Output channels	8	8	4
Control transport	Wired Ethernet	Wired Ethernet (WT32-ETH01)	Wi-Fi, or Ethernet via brain board
Ethernet implementation	W5500 dedicated controller on SPI — own MAC, PHY and hardware TCP/IP	ESP32 internal RMII MAC + external PHY — consumes 9 GPIO	Per brain board
Controller power	PoE from the switch	From the pixel supply	From the pixel supply
Reachable with pixel supply off	Yes	No	No
Pixel input range, one SKU	5–48 VDC	V4: 12–48 V (5 V support dropped) · V3: 5–24 V (discontinued)	5–24 VDC
Covers 5/12/24/36 V pixels	Yes, one board	No — requires choosing a version	Jumper-selectable up to 24 V; 36 V not supported
Input voltage selection	None — nothing to set wrong	Per board version	Jumper; vendor warns wrong setting "will fry the onboard components"
Input connector	Enclosed 57 A screw terminal	V4: spade connectors on a stud	Screw terminal
Continuous current, standard copper	30 A	Not published	15 A at 1 oz (30 A only with paid 2 oz upgrade)
Per-channel fuse rating	7.5 A nominal · 10 A max per output	5 A (V4) · 4 A hold / 8 A trip polyfuse (V3)	10 A max recommended
Fuses on outputs	8 — one per channel	8 — one per channel	5 fuses across 7 outputs (channels share)
Reverse-polarity protection	Active blocking — non-destructive, self-recovering	Not published	Parallel diode with fuse — sacrificial
Input transient clamp	Bidirectional TVS across the input	Not published	Not published
Data-line ESD protection	Per channel, sub-pF device	Series resistor only (33 Ω)	Not published
Power domain separation	3 pixel domains + isolated network domain	Not published	Single rail
DMX-512 / RS-485 output	4 ports, individually protected	Not published	Not published
Level shifter	8-channel, socketed DIP	Not published	4-channel, soldered
Conformal coating	Yes, applied at the fab	Not offered	Not offered
Fuse-status indicator	Per-channel LED	Blue "good fuse" LED per channel	Per board
Onboard USB programming	Via the module's own port	Onboard USB-C — required, the ESP is soldered down	Via the module's own port
Field-replaceable MCU	Yes — socketed module	No — ESP32-WROOM soldered to the board	Yes — socketed module
Fuse replacement	Mini blade — auto parts store, gas station, variety pack	Cartridge in a surface-mount holder — order online	ATO blade — auto parts store
Audio-reactive / mic support	DIY — headers fitted, spare GPIO available	DIY — wire an INMP441 to the H1 header (documented)	Analog audio input broken out
Temperature sensor	DIY — headers fitted, spare GPIO available	DIY — add headers DS18B20 and 4.7 kΩ	Optional onboard fast-die

Temperature sensor support	DIY — headers fitted, spare GPIO available	DIY — add header, US18B20 and 4.7 kΩ (documented)	Optional onboard footprint
Enclosure	Coming soon	None — 3D-print mounts published	None — bare board
Board size	123 × 129 mm — the cost of blade holders, a DIP socket and terminal clearance	~110 × 70 mm	~100 × 50 mm
Firmware	WLED / ESPHome / ESPixelStick — module-level, not board-level	Same	Same
Firmware image	Pre-built Ethernet image supplied	Pre-built image supplied; ESPixelStick needs the vendor hardware profile	Stock WLED builds
Ecosystem	New	Established, reviewed, xLights docs	Large, mature, well documented

A note on firmware comparisons. WLED, ESPHome, and ESPixelStick are properties of the ESP32 silicon, not of the carrier board — any of them will run on any of these products. What actually differs is which pre-built image ships, and how much GPIO the board's own hardware has already spent before your channels get any.

Where the others are the better choice, plainly. Bong69 ships a proven, well-reviewed Ethernet board with onboard USB-C programming, per-fuse status LEDs, documented audio-reactive and temperature-sensor support, xLights integration notes, published 3D-print mounts, and boards that arrive pre-configured and tested. QuinLED brings a deep peripheral breakout and years of community documentation. Neither ecosystem exists here yet. If you want established products, buy theirs.

Where NODE Backbone is the better choice. Everyone in this class can do Ethernet. Almost nobody powers the controller from it. PoE means one cable to the node, no second low-voltage feed into the enclosure, and a controller that stays on the network with the pixel supply dark — the difference between diagnosing a dead zone from your desk and diagnosing it from a ladder. Add one board covering 5/12/24/36 V pixels with every part in the power path well below its rating — no version to pick, an enclosed 57 A terminal instead of exposed lugs, and four levels of protection on separate copper.

And plainly: if one controller sits in a garage and works fine on Wi-Fi, buy NODE Mini and save the money. NODE Backbone earns its price at the scale where wireless stops being reliable — a threshold that arrives earlier than most people expect.

Mounting & Enclosure

No enclosure is included, and none is planned. This is the norm in this segment — bare board, mounted in whatever the installation already uses: a CG1500, a DIN rail, a weatherproof box, a plywood backer.

What NODE adds instead is **conformal coating, applied at the fabrication facility** with connectors, sockets, fuse clips, and switches masked. Competing boards in this class ship uncoated. A coated board in a vented enclosure tolerates condensation and humidity in a way a bare board does not, which matters for the seasonal outdoor installs this was built for.

Four corner mounting holes are provided. *Hole spacing and a downloadable mount STL: to be published at first article.*

Enclosure note specific to this variant: the Ethernet jack requires clearance and cable strain relief at the enclosure wall. Plan the entry before drilling.

Build & Assurance

- **Fabrication** — 4-layer, 1.6 mm, high-T_g substrate, ENIG finish, epoxy-filled and capped vias, IPC Class 2 workmanship
- **Assembly** — professional SMT assembly, flying-probe electrical test on 100% of boards
- **Traceability** — 2D barcode serial on every board; UL-recognized substrate marking
- **Coating** — conformal coating applied at the fab, with connectors, sockets, fuse clips, and switches masked
- **Design review** — every net verified against manufacturer datasheets; schematics independently reviewed by two automated analysis passes with human verification of every finding

Ordering

Item	Detail
Model	NODE Backbone
Configuration	Assembled, tested, conformal coated
Included	Controller board, level shifter installed, fuses installed
Not included	Enclosure, Waveshare ESP32-S3-ETH-POE module, 3-pin plug connectors, blade fuses beyond those fitted, pixel supply, Ethernet cable
Availability	Pre-release. The present build is a validation run — first-article hardware for bring-up, characterisation and field trial.
Price	TBD
Lead time	TBD
Warranty	TBD

NODE Backbone — Technical Specification

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Components are specified by function and rating. Specific manufacturers and part numbers are selected at build time and may change with availability; every substitution is held to the ratings stated here.

1. Power Architecture

Three pixel-side domains plus an independent network-powered logic domain:

Domain	Boundary	What it carries
Input (unprotected)	Screw terminal → ideal-diode stage	Raw supply. Assumed hostile. Transient clamp sits across this node.
Clean (protected)	Ideal-diode output → main fuse	Reverse-blocked, surge-clamped. Bulk capacitance lives here.
Distribution bus (fused)	Main fuse → 8 channel fuses	Every output taps this bus through its own fuse.
Logic (network-powered)	PoE module 5 V output → buffer and indicators	Independent of pixel supply state. Ground domain joined at one defined point.

2. Pixel Input & Protection

Parameter	Value	Notes
Pixel input voltage range	5 - 48 VDC	36 V nominal, 48 V absolute maximum
Input connector	Enclosed 2-pole screw terminal	57 A / 10 mm ² rated; 30 A design maximum = 53% of rating
Maximum continuous current	30 A	At standard 1 oz outer copper — no copper upgrade required
Main fuse	30 A mini blade, user-replaceable	Holder rated 30 A / 500 V
Reverse-polarity protection	Ideal-diode controller driving a high-side N-channel pass FET	FET held off under reverse input; no current path to load; non-sacrificial and self-recovering
Pass element	80 V N-channel MOSFET, low milliohm R _{DS(on)}	Rated above the input clamp voltage, not below it; current rating several times the 30 A design load
Input transient clamp	Bidirectional TVS, 1500 W, 48 V standoff	Conducts in neither direction at rated input; polarity-independent
Bulk capacitance	2 × 1000 µF, 63 V	On the clean rail, downstream of the ideal diode
Ideal-diode support	100 V charge-pump capacitor, 80 V hold-up capacitor	Per the controller's reference design
Polarity indication	Anti-parallel green/red LED pair	Green = correct, red = reversed; shared limit resistor sized for the full input range

Voltage headroom summary: 36 V nominal, 48 V absolute maximum. Every component in the power path is rated **63 V or above**; every semiconductor is **60 V or above**.

Derating against input voltage. Utilization of each component's own rating. **5 V, 12 V and 24 V are the mainstream pixel voltages** and are shown in bold:

Component	Rating	5 V	12 V	24 V	36 V	48 V
Bulk input capacitors	63 V	8 %	19 %	38 %	57 %	76 %
Channel output caps	80 V	6 %	15 %	30 %	45 %	60 %
Ideal-diode hold-up	80 V	6 %	15 %	30 %	45 %	60 %
HV-plane ceramics	100 V	5 %	12 %	24 %	36 %	48 %
Ideal-diode controller	65 V	8 %	18 %	37 %	55 %	74 %
Pass MOSFET V _{DS}	80 V	6 %	15 %	30 %	45 %	60 %

At the **36 V nominal** rating no part on the plane exceeds 60 % of its own limit. At the **48 V absolute maximum** nothing exceeds 80 %. On the 5 V, 12 V and 24 V strips that carry the overwhelming majority of installations, the entire plane runs under 40 %.

On this board, "power path" means *pixel* power path. The logic rail is a separate story — see Section 3.

3. Logic Rail

Parameter	Value	Notes
5 V source	Waveshare ESP32-S3-ETH-POE module	No onboard buck converter on this variant
Supply path	Module 5 V output → buffer, indicators	Independent of the pixel input rail
5 V rail clamp	Unidirectional TVS, 6 V standoff	Protects the PoE-derived rail
Grounding	Module ground domain merged to board ground at a single defined point	Isolation handled explicitly in layout
Behavior with pixel supply off	Controller remains powered and network-reachable	Configuration and diagnostics available without strip power

4. Outputs

Parameter	Value	Notes
Channels	8	Independent, individually fused — no shared fuses
Output connector	3-pole pluggable terminal, per channel	V+ / DATA / GND; 12 A rated → 63% at the 7.5 A fitted fuse, 83% at the 10 A maximum
Per-channel fuse	7.5 A mini blade fitted, 10 A maximum	25% of holder rating at 7.5 A, 33% at 10 A; user-replaceable
Level translation	Octal logic buffer, DIP-20 socket	Accepts either of two common octal families; 3.3 V drive against a 2.0 V V _{IH} = 1.3 V margin
Data series resistance	330 Ω, axial through-hole, per channel	Limits fault current back into the buffer
Data ESD protection	Sub-picofarad ESD device, per channel, connector side	0.4–0.55 pF junction capacitance; RC ≈ 181 ps — no measurable effect on WS281x bit timing
Data drive current	≈ 4 mA per channel	67% of the ±6 mA buffer rating
Buffer supply current	≈ 45 mA total	64% of the 70 mA device maximum
Per-channel decoupling	47 µF electrolytic 80 V + 100 nF ceramic 100 V	Local to each output connector

Channel activity indication	One LED per channel	Data activity on that channel
Fuse-status indication	One LED per channel, fed from the post-fuse node	Lit = that channel's fuse is intact and the bus is live

4b. DMX-512 / RS-485 Output

Parameter	Value	Notes
Ports	4	RJ45, T568B pairing
Transceivers	One RS-485 transceiver per port	Driver enables tied to a shared enable net; transmit data shared across all four
Receive	Port 1 only	The remaining three ports are transmit-only by design
Port protection	TVS array on the connector-side net, plus series resistance per line	Clamp sits ahead of the series limiting, in that order

5. MCU, Network & Firmware

Parameter	Value
Module	Waveshare ESP32-S3-ETH-POE (<i>not included</i>)
Mounting	Socketed — removable without desoldering
Network interface	Wired 10/100 Ethernet, W5500 controller on SPI
Ethernet architecture	Dedicated MAC + PHY + hardware TCP/IP offload; MCU cores not used for the stack
Power over Ethernet	IEEE 802.3af compliant, via the module's PoE daughterboard
Wireless	Wi-Fi available on module; wired Ethernet is the intended control path
Firmware	WLED — pre-built Ethernet image supplied (W5500 SPI configuration)
Supported LED protocols	WS2811, WS2812/B, WS2813, WS2815, WS2816, SK6812/RGBW, TM1814, and other single-wire families supported by WLED
Control protocols	Per WLED: E1.31/sACN, Art-Net, DDP, MQTT, HTTP/JSON API. DMX-512 over the onboard RS-485 ports
Additional I/O	Spare module GPIO broken out on fitted headers for I ² C and SPI peripherals, plus 3.3 V / 5 V / GND / TX / RX — no LED channels sacrificed
GPIO restriction	GPIO33–GPIO37 are consumed by the module's octal PSRAM and are unavailable; channel assignments avoid these pins by design

Integrator note: the module is an 802.3af powered device. Confirm the total PoE budget on your switch before populating it with many nodes.

6. Physical

Parameter	Value
PCB dimensions	123 × 129 mm
PCB thickness	1.6 mm
Layer count	4 — signal / ground plane / split power plane / signal
Copper weight	1 oz outer, 1 oz inner
Substrate	High-T _g 155 °C
Surface finish	ENIG, 1 μ" gold
Via treatment	Epoxy filled and capped
Soldermask	Blue
Mounting	4 × corner mounting holes, 3 mm
Mounting hole spacing	107.9 × 103.8 mm
Enclosure	Coming soon
Net weight	TBD — measure at first article
Conformal coating	Applied at the fab; connectors, sockets, fuse clips, and switches masked

7. Environmental & Compliance

Parameter	Value
Operating temperature	−20 °C to +60 °C (<i>design target; not qualification tested</i>)
Substrate rating	UL-recognized, JLC-1 marking
RoHS	Compliant
Workmanship	IPC Class 2
Electrical test	Flying-probe, 100% of boards
Traceability	2D barcode serial number, per board

8. Design Notes & Accepted Limitations

- Stated plainly rather than buried:
- **Not automotive qualified.** Protection targets transients credible in a residential or light-commercial installation. Not tested to ISO 7637 load-dump profiles; not intended for vehicle use.
 - **Worst-case clamp coordination.** At the TVS datasheet maximum clamping voltage under a full rated pulse, the voltage presented to downstream components exceeds their ratings. That condition requires a pulse energy source which does not exist in a mains-fed low-voltage installation, and is

accepted as non-credible for the intended deployment. At realistic clamp currents the entire downstream chain retains margin.

- **Over-voltage beyond specification is not survivable.** A supply above the rated input range will conduct the input TVS and open the main fuse. Sacrificial by design — the board is protected, the TVS and fuse are consumables.
- **No onboard 5 V regulator.** Logic power comes exclusively from the module. An unpowered Ethernet drop means an unpowered controller regardless of pixel supply state.
- **PoE budget is the installer's responsibility.** Node count per switch is bounded by the switch's total PoE power budget, not by this board.
- **Load capability.** Pixel power comes directly from the distribution bus; total system current is bounded by the 30 A main fuse and the user's supply.

Wateefy Electronics · NODE Backbone · Specifications subject to change.